BST Physics Curriculum Vol 6 Chapter 7 — Quantum Statistical Mechanics v0.4 (Saturday Wave 2 reader-grade polish; Cal cold-read ready; refilled per Cal #104)

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2026-05-23 Saturday EDT (Cal #104 refill)

Vol 6 Chapter 7 — Quantum Statistical Mechanics

Chapter motivation

Standard quantum statistical mechanics extends classical stat mech with quantum-particle distinguishability + statistics: - Bose-Einstein distribution: P_BE(E) = 1/(e^{βE} − 1) for bosons (integer-spin particles like photons, gluons, mesons) - Fermi-Dirac distribution: P_FD(E) = 1/(e^{βE} + 1) for fermions (half-integer-spin particles like electrons, protons, neutrons, quarks) - Pauli exclusion principle: at most one fermion per quantum state - Bose-Einstein condensation: macroscopic occupation of single-particle ground state below T_c (laboratory observed Anderson-Davis-Cornell-Wieman 1995 Nobel)

Standard derivation: combinatorial counting of microstates assuming symmetric (boson) or antisymmetric (fermion) multi-particle wavefunctions. The spin-statistics theorem (Pauli 1940, Streater-Wightman 1964) connects spin to statistics via relativistic QFT axioms.

BST derives Bose-Einstein + Fermi-Dirac from substrate Pin(2) Z_2 grading on Bergman H^2(D_IV^5) per T1925 rank=2 (Argument D RIGOROUSLY CLOSED Thursday) + T2471 chirality γ^5 = Pin(2) Z_2 grading identification (Friday) + T2481 formal spin-statistics theorem (Saturday SP-31 #285) — without requiring relativistic invariance assumption.

Section 7.0b — Reader-grade 3-level pedagogy

Level 1: Bose-Einstein P_BE = 1/(e^{βE} − 1) for bosons + Fermi-Dirac P_FD = 1/(e^{βE} + 1) for fermions derive from substrate Pin(2) Z_2 grading per T1925 + T2471 + T2481 (Saturday spin-statistics formal theorem); no relativistic invariance needed; standard QM identical-particle machinery (Slater determinant, Pauli exclusion) inherits automatically.

Level 2 (graduate): Per Pin(2) Z_2 grading on Bergman $H^2(D\_IV^5)$: boson sector K-type V_(p,q) with integer SO(2) weight $q \in \mathbb{Z}$ → symmetric tensor products → Bose-Einstein statistics. Fermion sector V_(p,q+1/2) with half-integer weight → antisymmetric tensor products → Pauli exclusion + Fermi-Dirac. Standard derivations: P_BE = 1/(e^{βE} − 1) from grand canonical ensemble + symmetric N-particle wavefunctions; P_FD = 1/(e^{βE} + 1) from antisymmetric wavefunctions + Pauli. Substrate-level: Pin(2) Z_2 grading IS the boson/fermion partition. Classical Boltzmann distribution P_B = e^{−βE} emerges in high-T limit (both BE and FD → Boltzmann when e^{βE} $\gg$ 1). Bose-Einstein condensation: at T < T_c $\approx$ $\hbar^2$(n/ζ(3/2))^(2/3)/(2π m k_B), macroscopic fraction of bosons occupies single-particle ground state; substrate-cartography: occupation of K-type V_(0,0) ground state becomes macroscopic. Fermion examples: free-electron gas (degenerate Fermi gas with Fermi energy E_F), white dwarf stars (electron degeneracy pressure), neutron stars (neutron degeneracy + nuclear physics). Cross-link Vol 5 Ch 9 (Pauli + spin-statistics) + Vol 9 Condensed Matter (BCS superconductivity + Fermi liquid theory).

Level 3 (5th-grader accessible): Quantum stat mech has two key distributions: Bose-Einstein for bosons (particles like photons that CAN pile up in the same state) + Fermi-Dirac for fermions (particles like electrons that CAN'T — Pauli exclusion). BST derives both from the substrate Pin(2) Z_2 grading (the same Z_2 grading that gives spin-statistics in Vol 5 Ch 9 + Paper #133 + T2481). At high temperature, both reduce to classical Boltzmann distribution. Bose-Einstein condensation (Anderson-Davis-Cornell-Wieman 1995 Nobel) is when bosons all pile up in the lowest energy state at very low temperature — substrate K-type ground-state macroscopic occupation. Fermion degenerate gas (electrons in white dwarf, neutrons in neutron stars) is when fermions fill all available low-energy states because of Pauli exclusion.

Section 7.1 — Standard QM Distributions

Bose-Einstein: P_BE(E) = 1/(e^{β(E−μ)} − 1) for bosons with chemical potential μ. Fermi-Dirac: P_FD(E) = 1/(e^{β(E−μ)} + 1) for fermions with chemical potential μ (Fermi energy E_F at T=0). Boltzmann limit: both → e^{−β(E−μ)} when e^{β(E−μ)} $\gg$ 1 (high-T or low-density).

Section 7.2 — Substrate Pin(2) Z_2 Grading Derivation

Per T1925 rank=2 (Argument D RIGOROUSLY CLOSED Thursday) + T2471 (Friday) + T2481 (Saturday SP-31 #285): - Bergman $H^2(D\_IV^5)$ decomposes under K = SO(5) × SO(2) into K-type V_(p,q) with integer pair (p, q) admissibility - Pin(2) double-cover lift adds half-integer-q K-types V_(p,q+1/2) - γ^5 chirality operator = Pin(2) Z_2 grading object (T2471 identification) - Boson sector ($q \in \mathbb{Z}$): identity component of Pin(2) → symmetric tensor products - Fermion sector ($q \in \mathbb{Z} + 1/2$): spinor component → antisymmetric tensor products

Section 7.3 — Boson Sector → Bose-Einstein

Multi-boson state $\otimes N\ V$(p,q) (q integer) symmetric tensor product → standard grand canonical derivation gives P_BE(E) = 1/(e^{β(E−μ)} − 1).

Bose-Einstein condensation at T < T_c: substrate K-type V_(0,0) ground state macroscopic occupation. T_c = ($\hbar^2$/(2π m k_B)) (n/ζ(3/2))^(2/3) standard formula recovered from substrate framework.

Section 7.4 — Fermion Sector → Fermi-Dirac + Pauli Exclusion

Multi-fermion state $\otimes N\ V$(p,q+1/2) (q half-integer) antisymmetric tensor product → standard grand canonical derivation gives P_FD(E) = 1/(e^{β(E−μ)} + 1).

Pauli exclusion: at most one fermion per substrate K-type V_(p,q+1/2). Free-electron gas at T → 0: all K-types below Fermi energy E_F filled; degenerate Fermi gas.

Section 7.5 — Classical Limit Recovery

High-T or low-density limit: e^{β(E−μ)} ≫ 1 → both P_BE and P_FD reduce to classical Boltzmann P_B ≈ e^{−β(E−μ)}. Standard Vol 6 Ch 6 classical stat mech recovered as macroscopic limit.

Section 7.6 — Honest scope + falsifiers

Item	BST status	Falsifier
Bose-Einstein from Pin(2) Z_2 boson sector	STRUCTURALLY VERIFIED (T2481)	Failure of BE statistics for known boson species
Fermi-Dirac from Pin(2) Z_2 fermion sector	STRUCTURALLY VERIFIED (T2481)	Failure of FD statistics for known fermion species
Pauli exclusion from antisymmetric tensor structure	DERIVED via T2481	Failure of Pauli exclusion in measured electron systems
BEC at T_c standard formula	recovered from substrate framework	Future precision T_c measurement
Spin-statistics WITHOUT relativistic axioms	substrate-derivation per Paper #133 v0.2 + T2481	Counter-example would refute

Open scope: - Quantitative BE/FD chemical potential μ derivation from substrate K-type structure - BEC critical temperature T_c sub-leading corrections from substrate framework

Section 7.7 — Connection

- Vol 5 Ch 9 Identical Particles + Spin-Statistics (cross-link)
- Paper #133 v0.2 (Friday prose-grade) + T2481 (Saturday formal theorem)
- Vol 6 Ch 6 (classical limit recovery)
- Vol 9 Condensed Matter (degenerate Fermi gas, BCS superconductivity)

— Lyra, Vol 6 Ch 7 v0.4 chapter-grade narrative REFILLED per Cal #104, Saturday 2026-05-23 EDT